

13

Quadratic Form

Having Both Magnitude and Direction

This chapter begins with the foundational concept of "positive definiteness," helping you build a geometric intuition for positive definite, positive semidefinite, negative definite, negative semidefinite, and indefinite matrices. We will also learn how to determine whether a matrix is positive definite, and reveal the close connection between positive definite matrices, inner products, and quadratic forms. Building on this, we introduce two commonly used matrix decomposition methods: Cholesky decomposition and LDL decomposition. Next, we will take a deep dive into the Rayleigh quotient, whose numerator and denominator are both quadratic forms. Finally, this chapter introduces several common distance metrics, including the Mahalanobis distance — the origin of the rotated ellipses you've seen frequently in earlier sections of this book.

13.1 Positive Definiteness



Core skills you will master in this section:

- ▶ The definitions of positive definite, positive semidefinite, negative definite, negative semidefinite, and indefinite matrices.
- ▶ Positive definite: an upward-opening paraboloid.
- ▶ Positive semidefinite: a valley surface.
- ▶ Negative definite: a downward-opening paraboloid.
- ▶ Negative semidefinite: a ridge surface.
- ▶ Indefinite matrix: a saddle surface.
- ▶ Judging a matrix's positive definiteness from the perspective of its eigenvalues.

Quadratic Forms

Simply put, a quadratic form is a degree-two polynomial in the vector $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$. Its general form is as follows

$$f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

Here, the matrix $\mathbf{A}$ is a real symmetric matrix.

As shown in Figure 1, the matrix multiplication above yields a 1×1 matrix, equivalent to a scalar.

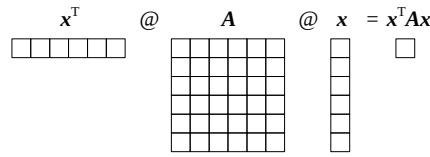


Figure 1. The Matrix Multiplication Corresponding to a Quadratic Form

The equation above is essentially a multivariate quadratic function.

For example, given the following symmetric matrix A

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

We easily obtain the bivariate quadratic function $f(\mathbf{x})$

$$f(\mathbf{x}) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2x_1^2 + 2x_1x_2 + 3x_2^2$$

The rest of this section will explain quadratic forms and positive definiteness from the perspective of bivariate functions.



Beauty of Visualization explains how to use the "tofu-slicing" technique to visualize three-variable quadratic forms.

Positive Definiteness

Based on the sign properties of a quadratic form, we can classify it into five categories.

When $\mathbf{x} \neq \mathbf{0}$ ($\mathbf{x}$ is a nonzero column vector), if the following holds:

$$\mathbf{x}^T A \mathbf{x} > 0$$

The matrix A is a positive definite matrix.

When $\mathbf{x} \neq \mathbf{0}$, if

$$\mathbf{x}^T A \mathbf{x} \geq 0$$

The matrix A is a positive semidefinite matrix.

When $\mathbf{x} \neq \mathbf{0}$, if

$$\mathbf{x}^T A \mathbf{x} < 0$$

The matrix A is a negative definite matrix.

When $\mathbf{x} \neq \mathbf{0}$, if

$$\mathbf{x}^T A \mathbf{x} \leq 0$$

The matrix A is a negative semidefinite matrix.

The matrix A does not fall into any of the above cases, A is an indefinite matrix.

A Geometric Perspective on Positive Definiteness

Given the following 2×2 real symmetric matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

Construct the following bivariate function $y = f(x_1, x_2)$:

$$y = f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = ax_1^2 + 2bx_1x_2 + cx_2^2$$

In three-dimensional orthogonal space, as the matrix $\mathbf{A}_{2 \times 2}$'s positive definiteness varies, the surface corresponding to $y = f(x_1, x_2)$ takes on different shapes:

- ◀ When $\mathbf{A}_{2 \times 2}$ is a positive definite matrix, $y = f(x_1, x_2)$ is an upward-opening paraboloid;
- ◀ When $\mathbf{A}_{2 \times 2}$ is a positive semidefinite matrix, $y = f(x_1, x_2)$ is a valley surface;
- ◀ When $\mathbf{A}_{2 \times 2}$ is a negative definite matrix, $y = f(x_1, x_2)$ is a downward-opening paraboloid;
- ◀ When $\mathbf{A}_{2 \times 2}$ is a negative semidefinite matrix, $y = f(x_1, x_2)$ is a ridge surface;
- ◀ When $\mathbf{A}_{2 \times 2}$ is indefinite, $y = f(x_1, x_2)$ is a saddle surface, also called a hyperbolic paraboloid.

Let's dive into the details below.

Positive Definite

The matrix $\mathbf{A}_{2 \times 2}$ being positive definite means that $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A}_{2 \times 2} \mathbf{x}$ is an upward-opening paraboloid, shaped like a bowl.

Let's start with a simple example. If the matrix $\mathbf{A}$ is the 2×2 identity matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The identity matrix is clearly positive definite.

Construct the following bivariate function $f(x_1, x_2)$:

$$f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + x_2^2$$

The surface and contour lines of the quadratic function are shown in Figure 2.

As shown in Figure 2 (b), when $x_1^2 + x_2^2 = c$ ($c > 0$), the contour lines are perfect circles.

It is easy to see that only when $x_1 = 0$ and $x_2 = 0$, that is, $\mathbf{x} = \mathbf{0}$, does $y = f(x_1, x_2) = 0$.

$\mathbf{x}$ can be any nonzero vector; the expression above is always greater than 0.

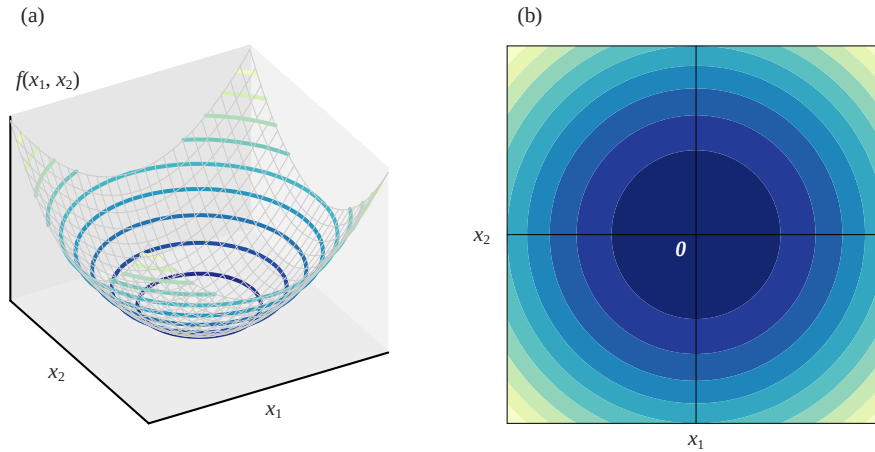


Figure 2. An Upward-Opening Circular Paraboloid

It is easy to find that, in the equation above, $\mathbf{A}$'s eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 1$, with the corresponding eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Compute the rank of matrix $\mathbf{A}$: $\text{rank}(\mathbf{A}) = 2$.

Let's look at another 2×2 positive definite matrix example. The specific values of matrix $\mathbf{A}$ are as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Similarly, construct the bivariate function $f(x_1, x_2)$

$$f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + 2x_2^2$$

For the expression above, only when $x_1 = 0$ and $x_2 = 0$ does $y = f(x_1, x_2) = 0$.

Figure 3(a) shows the upward-opening elliptical paraboloid corresponding to the equation above; as shown in Figure 3(b), the contour lines are a series of ellipses.

It is easy to find that $\mathbf{A}$'s eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$, with the corresponding eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

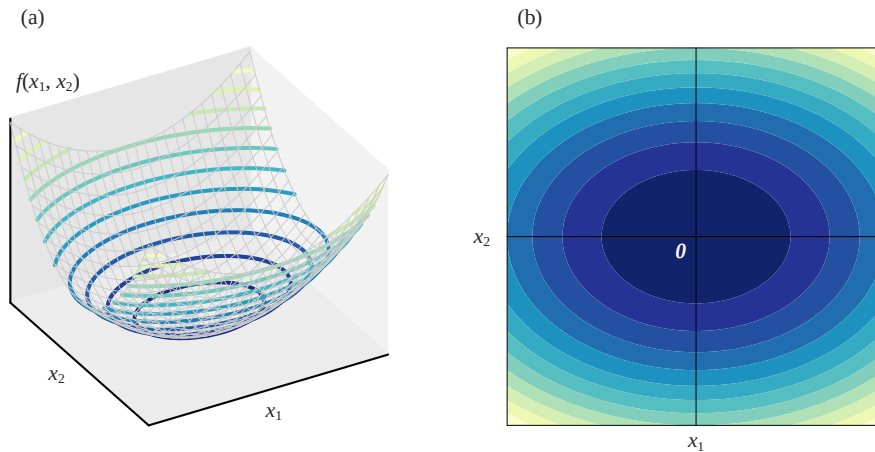


Figure 3. An Upward-Opening Elliptical Paraboloid

Now let's look at a case involving a rotated ellipse. The square matrix $\mathbf{A}$ is specifically as follows:

$$\mathbf{A} = \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{bmatrix}$$

Construct the bivariate function $f(x_1, x_2)$:

$$f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1.5x_1^2 + x_1x_2 + 1.5x_2^2$$

Similarly, only when $x_1 = 0$ and $x_2 = 0$ does $f(x_1, x_2) = 0$.

Through computation, we find that $\mathbf{A}$'s eigenvalues are also $\lambda_1 = 1$ and $\lambda_2 = 2$. The eigenvalue decomposition (spectral decomposition) of the equation above is

$$\mathbf{A} = \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T}$$

As shown in Figure 4 (b), the orthogonal matrix $\mathbf{V}$'s column vectors form the orthonormal basis $[\mathbf{v}_1, \mathbf{v}_2]$, which straightens the ellipse.

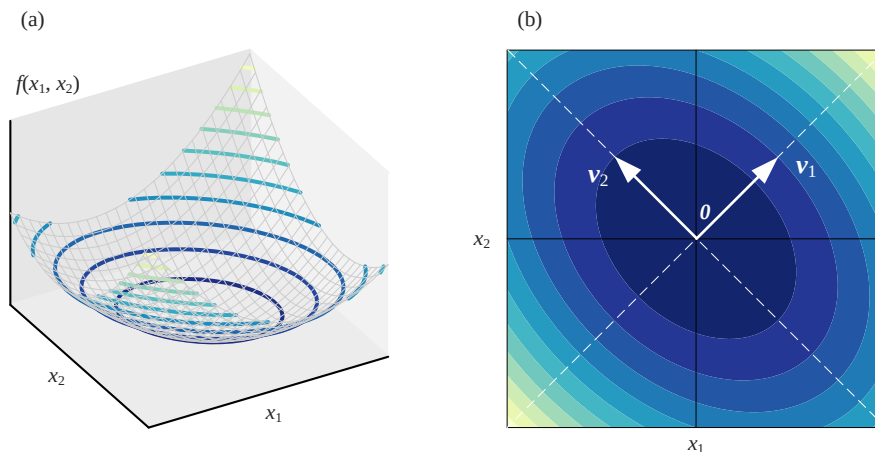


Figure 4. An Upward-Opening Rotated Elliptical Paraboloid

Positive Semidefinite

The matrix $\mathbf{A}_{2 \times 2}$ being positive semidefinite means that $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A}_{2 \times 2} \mathbf{x}$ is an upward-opening valley surface.

Let's look at the first example. The values of matrix $\mathbf{A}$ are as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

It is easy to determine that $\text{rank}(\mathbf{A}) = 1$, and the determinant is 0.

Construct the following bivariate function $f(x_1, x_2)$:

$$f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2$$

$x_1 = 0$, regardless of the value of x_2 , the expression above equals 0. In other words, besides the zero vector $\mathbf{0}$, there are countless other vectors that make the expression above equal 0; this is exactly why, in matrix $\mathbf{A}$ is called positive semidefinite.

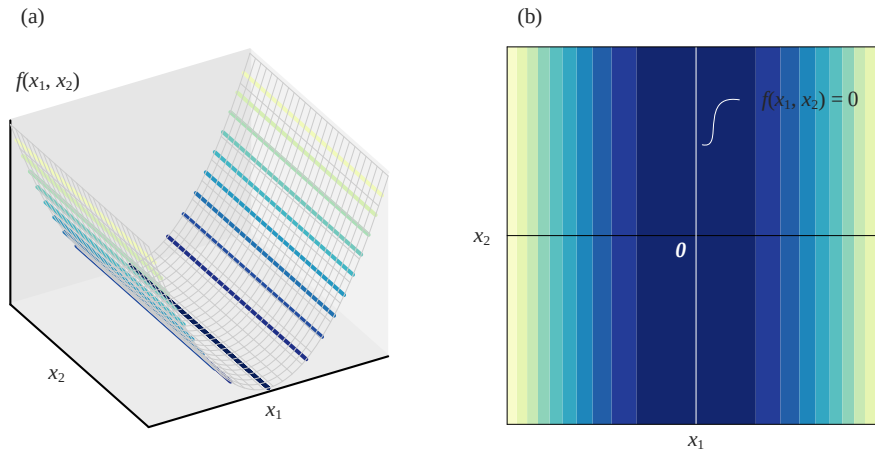


Figure 5. A Valley Surface

In the expression below, matrix $\mathbf{A}$ is also positive semidefinite:

$$\mathbf{A} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

The matrix $\mathbf{A}$'s determinant is 0, and its rank is 1.

Construct the function $y = f(x_1, x_2)$:

$$y = f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.5x_1^2 - x_1x_2 + 0.5x_2^2$$

Completing the square in the equation above gives:

$$f(x_1, x_2) = 0.5x_1^2 - x_1x_2 + 0.5x_2^2 = \frac{1}{2}(x_1 - x_2)^2$$

It is easy to see that, besides the zero vector $\mathbf{0}$, any vector lying on the line $x_1 = x_2$ also makes $y = f(x_1, x_2)$ equal 0.

In the equation above, matrix $\mathbf{A}$'s eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = 1$, with the corresponding eigenvectors as follows:

$$\mathbf{v}_1 = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

In the equation above, matrix $\mathbf{A}$'s eigenvalue decomposition (spectral decomposition) is

$$\mathbf{A} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T}$$

Figure 6 shows the rotated valley surface corresponding to the equation above.

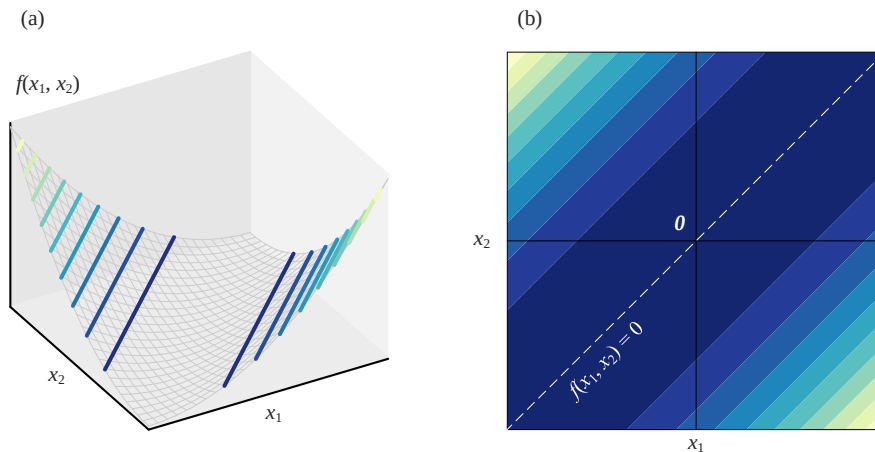


Figure 6. A Rotated Valley Surface

Negative Definite

The matrix $\mathbf{A}_{2 \times 2}$ being negative definite means that $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A}_{2 \times 2} \mathbf{x}$ is a downward-opening paraboloid.

The simplest negative definite matrix is the negated identity matrix, that is, $-\mathbf{I}$. $-\mathbf{I}$'s eigenvalues are all -1 .

Figure 7 shows the downward-opening circular paraboloid corresponding to $-\mathbf{I}$.



Please write out the bivariate function corresponding to Figure 7.

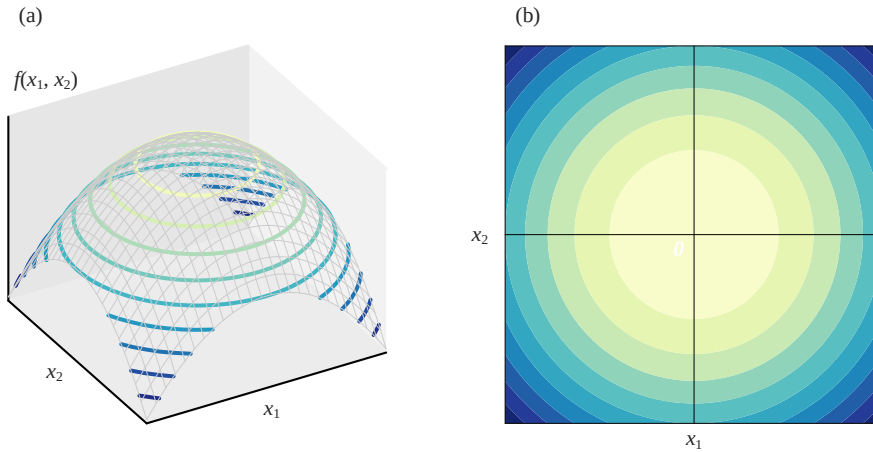


Figure 7. A Downward-Opening Circular Paraboloid

Let's also use a 2×2 matrix to discuss negative definiteness. The following matrix $\mathbf{A}$ is negative definite:

$$\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

Construct the following bivariate function $y = f(x_1, x_2)$:

$$y = f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -x_1^2 - 2x_2^2$$

As shown in Figure 8, it is easy to see that only when $x_1 = 0$ and $x_2 = 0$ does $y = f(x_1, x_2) = 0$.

It is very easy to find that $\mathbf{A}$'s eigenvalues are $\lambda_1 = -2$ and $\lambda_2 = -1$, with the corresponding eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

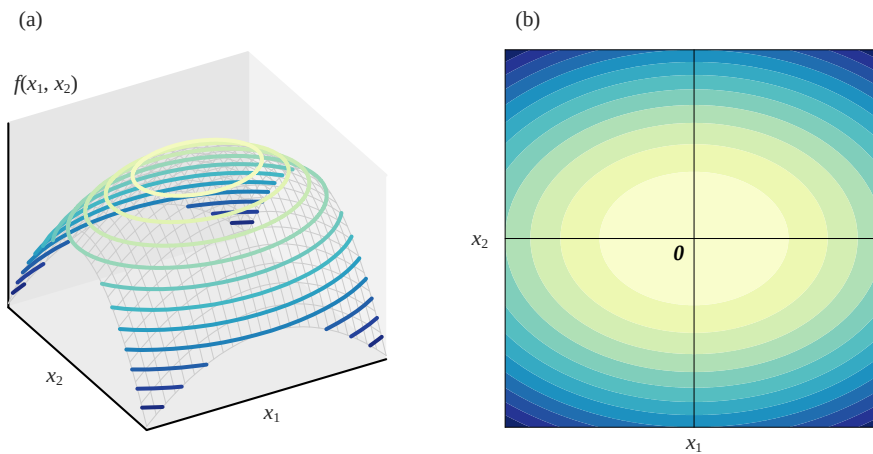


Figure 8. A Downward-Opening Elliptical Paraboloid

The following negative definite matrix $\mathbf{A}$

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$$\mathbf{A} = \begin{bmatrix} -1.5 & -0.5 \\ -0.5 & -1.5 \end{bmatrix}$$

corresponds to a downward-opening rotated elliptical paraboloid.

? Please compute the determinant and rank of the matrix above.

Construct the following bivariate function $y = f(x_1, x_2)$:

$$f(x_1, x_2) = -1.5x_1^2 - x_1x_2 - 1.5x_2^2$$

In the equation above, matrix $\mathbf{A}$'s eigenvalue decomposition is

$$\mathbf{A} = \begin{bmatrix} -1.5 & -0.5 \\ -0.5 & -1.5 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T}$$

Both eigenvalues are less than 0. The equation above also has rotated-ellipse contour lines. As shown in Figure 9 (b), the orthogonal matrix $\mathbf{V}$ is the orthonormal basis that straightens the rotated ellipse.

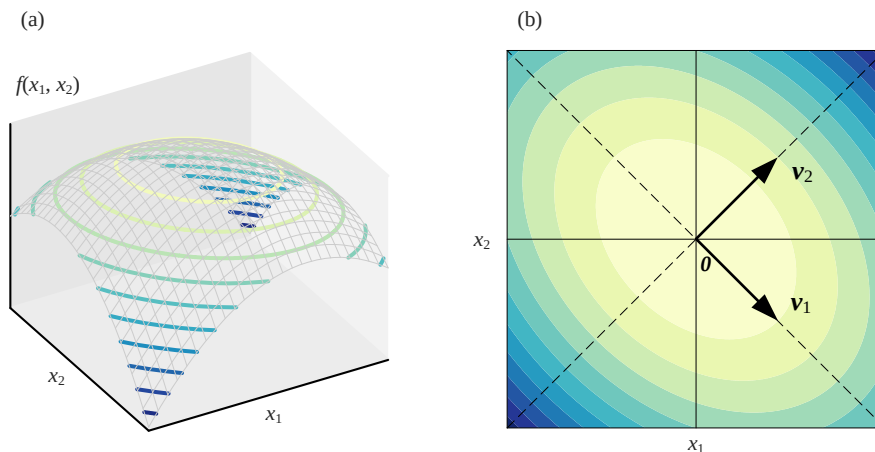


Figure 9. A Downward-Opening Rotated Elliptical Paraboloid

Negative Semidefinite

The matrix $\mathbf{A}_{2 \times 2}$ being negative semidefinite means that $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A}_{2 \times 2} \mathbf{x}$ is a downward-opening ridge surface.

Let's look at an example of a negative semidefinite matrix. The values of matrix $\mathbf{A}$ are as follows:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$$

Construct $y = f(x_1, x_2)$:

$$y = f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -x_2^2$$

$x_2 = 0$, x_1 can take any value, and the expression above equals 0. The matrix $\mathbf{A}$ has rank 1, $\text{rank}(\mathbf{A}) = 1$.

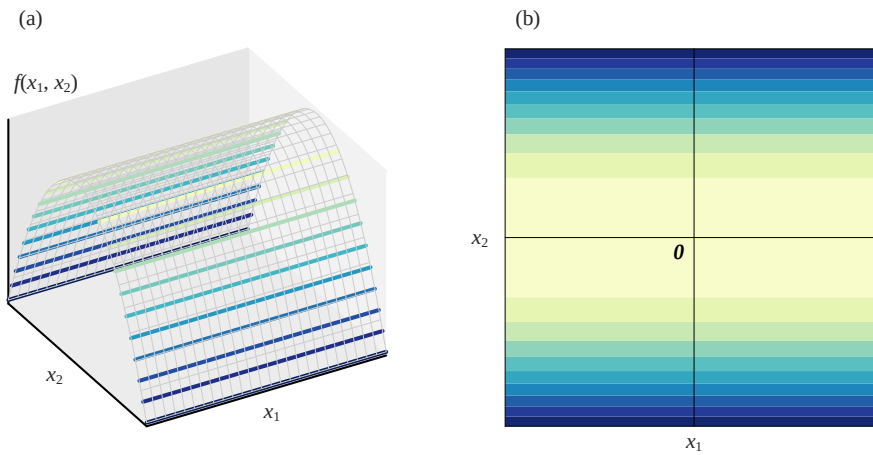


Figure 10. A Ridge Surface

Let's look at another example. The following negative semidefinite matrix

$$\mathbf{A} = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

? Please compute the determinant and rank of the matrix above.

The corresponding bivariate function is

$$f(x_1, x_2) = -0.5x_1^2 + x_1x_2 - 0.5x_2^2$$

Figure 11 shows the surface and contour lines corresponding to the equation above.

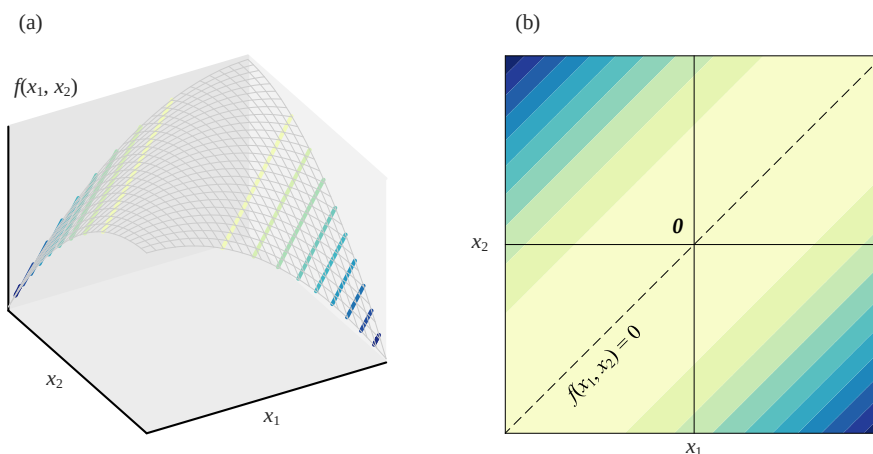


Figure 11. A Rotated Ridge Surface

Indefinite

The matrix $\mathbf{A}_{2 \times 2}$ being indefinite means that $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A}_{2 \times 2} \mathbf{x}$ is a saddle surface, with $(0, 0)$ as the saddle point. $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A}_{2 \times 2} \mathbf{x}$'s sign is indeterminate.

For example, $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Construct the function $f(x_1, x_2)$:

$$f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 - x_2^2$$

We find that matrix $\mathbf{A}$'s corresponding eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 1$, with the corresponding eigenvectors as follows:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Figure 12 shows the surface corresponding to $y = f(x_1, x_2)$.

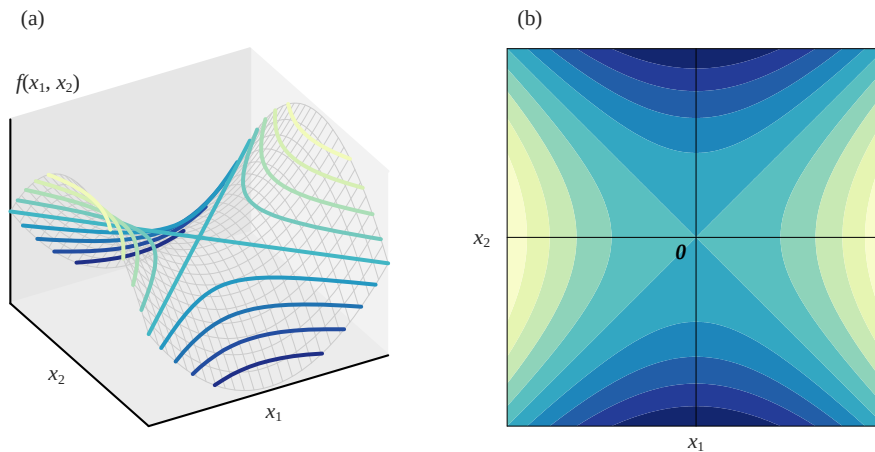


Figure 12. A Saddle Surface

When $f(x_1, x_2) \neq 0$, the surface's contour lines are hyperbolas.

Figure 12 shows that the saddle surface's center C is neither a local minimum nor a local maximum; in Figure 12, the center point of the saddle surface is called the saddle point.

Figure 12's saddle surface, rotated clockwise by 45° , gives the surface in Figure 13. Figure 13's surface corresponds to the matrix $\mathbf{A}$, as follows:

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

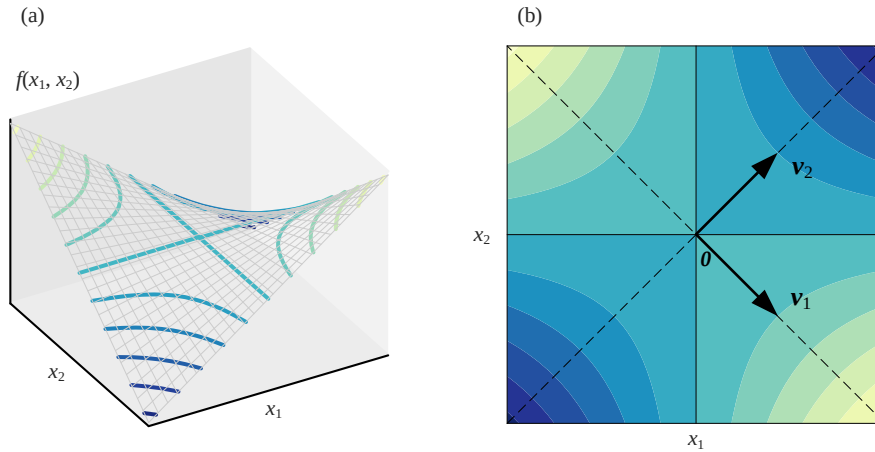


Figure 13. A Rotated Saddle Surface

Construct the following bivariate function $y = f(x_1, x_2)$:

$$y = f(x_1, x_2) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -2x_1x_2$$

When $f(x_1, x_2)$ takes a nonzero value, the expression above is equivalent to an inverse-proportion function.

In the equation above, matrix $\mathbf{A}$'s eigenvalue decomposition is

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T}$$

Eigenvalue Decomposition

Performing eigenvalue decomposition on the real symmetric matrix $\mathbf{A}$ gives:

$$\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$$

Substituting the equation above into $\mathbf{x}^T \mathbf{A} \mathbf{x}$, we get:

$$\begin{aligned} \mathbf{x}^T \mathbf{A} \mathbf{x} &= \mathbf{x}^T \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \mathbf{x} \\ &= \left(\mathbf{V}^T \mathbf{x} \right)^T \mathbf{\Lambda} \left(\mathbf{V}^T \mathbf{x} \right) \end{aligned}$$

Let:

$$\mathbf{z} = \mathbf{V}^T \mathbf{x}$$

The equation above can be written as:

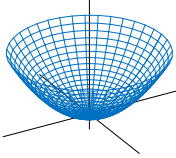
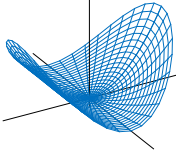
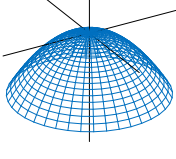
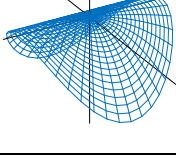
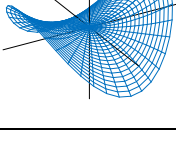
$$\begin{aligned} \mathbf{x}^T \mathbf{A} \mathbf{x} &= \mathbf{z}^T \mathbf{\Lambda} \mathbf{z} \\ &= \lambda_1 z_1^2 + \lambda_2 z_2^2 + \cdots + \lambda_D z_D^2 = \sum_{j=1}^D \lambda_j z_j^2 \end{aligned}$$

When the eigenvalues in the equation above are all positive, unless $z_1, z_2, \dots, z_n$ are all 0 (that is, $\mathbf{z}$ is the zero vector), the expression above is greater than 0, in which case $\mathbf{A}$ is positive definite.

Table 1 summarizes the surface shapes corresponding to matrix A 's different positive-definiteness conditions. Simply put,

- ▶ If A 's eigenvalues are all positive, A is positive definite.
- ▶ If A 's eigenvalues are nonnegative (positive or 0), A is positive semidefinite.
- ▶ If A 's eigenvalues are all negative, then matrix A is negative definite.
- ▶ If A 's eigenvalues are nonpositive (negative or 0), then matrix A is negative semidefinite.
- ▶ If A 's eigenvalues include both positive and negative values, then matrix A is indefinite.

Table 1. The Geometric Meaning of Positive Definiteness

$A_{n \times n}$	Eigenvalues	$x^T A x$'s Quadratic-Form Shape
$A_{n \times n}$ is positive definite $x^T A x > 0, x \neq 0$	All n eigenvalues are positive	
$A_{n \times n}$ is positive semidefinite, with rank r $x^T A x \geq 0, x \neq 0$	r eigenvalues are positive and $n - r$ eigenvalues are 0	
$A_{n \times n}$ is negative definite $x^T A x < 0$	All n eigenvalues are negative	
$A_{n \times n}$ is negative semidefinite, with rank r $x^T A x \leq 0$	r eigenvalues are negative and $n - r$ eigenvalues are 0	
$A_{n \times n}$ is indefinite	The signs of the eigenvalues are indeterminate (mixed positive and negative)	



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please determine the positive definiteness of the following square matrices.

▶ $\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$

▶ $\begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$

Q2. Please indicate the surface shape and contour-line characteristics of the quadratic form corresponding to each matrix in Q1.

Q3. Please write the following quadratic polynomial in the form $\mathbf{x}^T \mathbf{A} \mathbf{x}$, and indicate the surface shape and $\mathbf{A}$'s positive definiteness

▶ $-x_1^2 + x_2^2$

▶ $-x_1^2 + x_2^2 + 2x_1x_2$

▶ x_2^2

▶ $x_1^2 + x_2^2$

▶ $2x_1x_2$

▶ $2x_1^2 + 2x_2^2$

▶ $-2x_1x_2$

▶ $2x_1^2 + 2x_2^2 - 2x_1x_2$

▶ $-x_1^2$